



Commissioning a Solid-State Driven Accelerator

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Background + Motivation

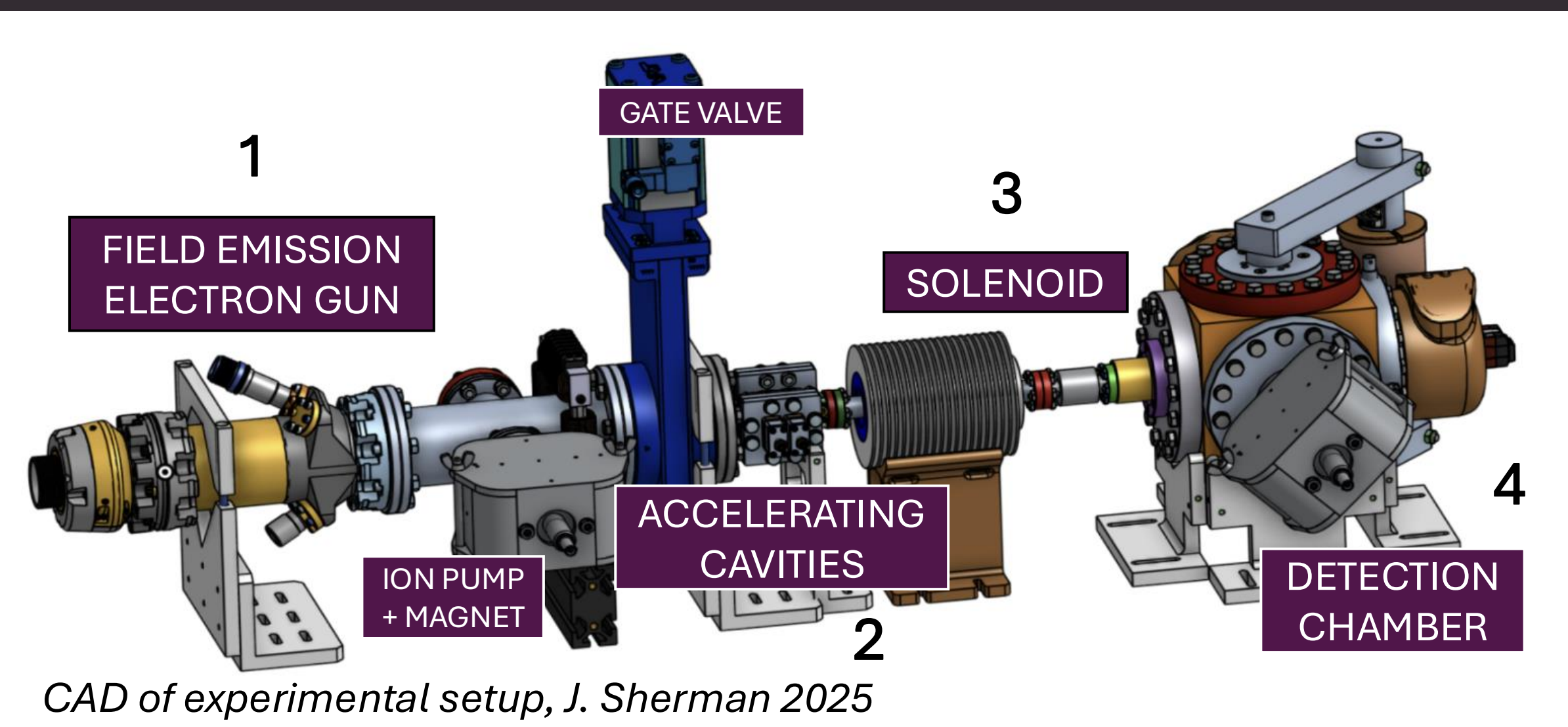
Particle accelerators output highly energized beams of particles with wide scientific applications across fields. Compact, energy-efficient designs have become increasingly important. Our lab has been commissioning a table-top solid-state-driven linear accelerator with an interest in electron microscopy.

Radio frequency (RF) accelerators use cavities to contain oscillating electromagnetic fields to accelerate particles and direct the beam path. Energy is transferred from electromagnetic fields to charged particles, which experiences the Lorentz's force:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = m\mathbf{a}$$

- The electric field accelerates the particles in the field direction
- The force due to the magnetic field is always perpendicular, so it only changes the direction of momentum.

Standard linacs require high-voltage klystrons as power sources for these cavities, which are complex and costly for smaller institutions. Instead, this setup powers each cavity with a solid-state amplifier board, which lowers the cost and scale of our system.



CAD of experimental setup, J. Sherman 2025

The linac setup includes a particle source, accelerating cavities, an electromagnet solenoid, and a detection chamber.

- The electron gun emits an electron beam, which is guided through the system via steering magnets.
- The four cavities accelerate the electrons.
- The solenoid focuses the beam on the axis.
- In the detection chamber, electrons are displaced by a dipole magnet energy spectrometer, then converted into photons via a phosphor screen, and captured by camera for detailed analysis.

Objectives

This summer our lab has three focuses: electron gun testing, dipole characterisation, and camera setup.

Electron Gun Testing:

- Test emission under a stable Ultra High-Vacuum (UHV) environment
- Achieve a steady beam and stable system by mapping the relation between emitted current and source voltage
- Eventually, measure an energy spread by connecting the gun to the detection chamber without cavities

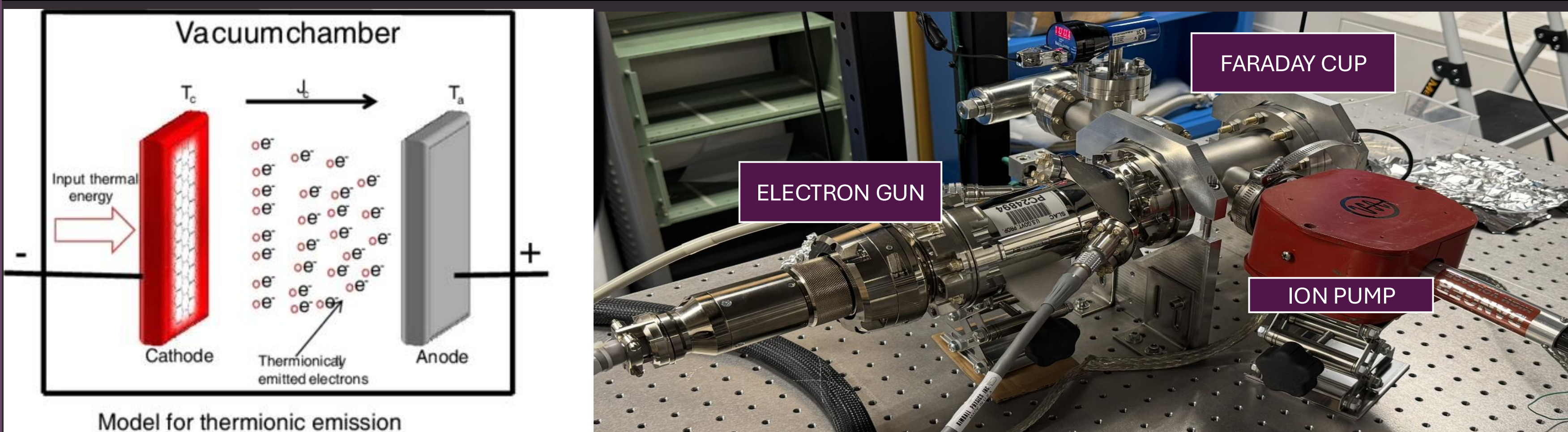
Dipole Characterisation (Coco): Measure the magnetic field (B) of the dipole magnet in the energy spectrometer.

- Design a measurement set based on a Hall effect sensor chip
- Mount sensor on a 3D motion system with motors and micrometers
- Implement data acquisition and processing with Arduino and Python

Camera Setup (Wamda):

- Write continuous acquisition code for the camera that can be synchronized with the RF system.
- Develop image processing program that identifies signal frames, classifies beam characteristics, uses data from camera to calculate energy spread via pixel intensity

Thermionic Emission Gun Setup



Liang, Shi-Jun & Ang, L.K. & Chen, G. (2014). Current gun Setup

Our setup uses a commercial DC electron gun, Kimball Physics EMG-4212 Electron Gun, and supplies high voltage potential from a 30 kV Kimball EGPS-4212 Power Supply.

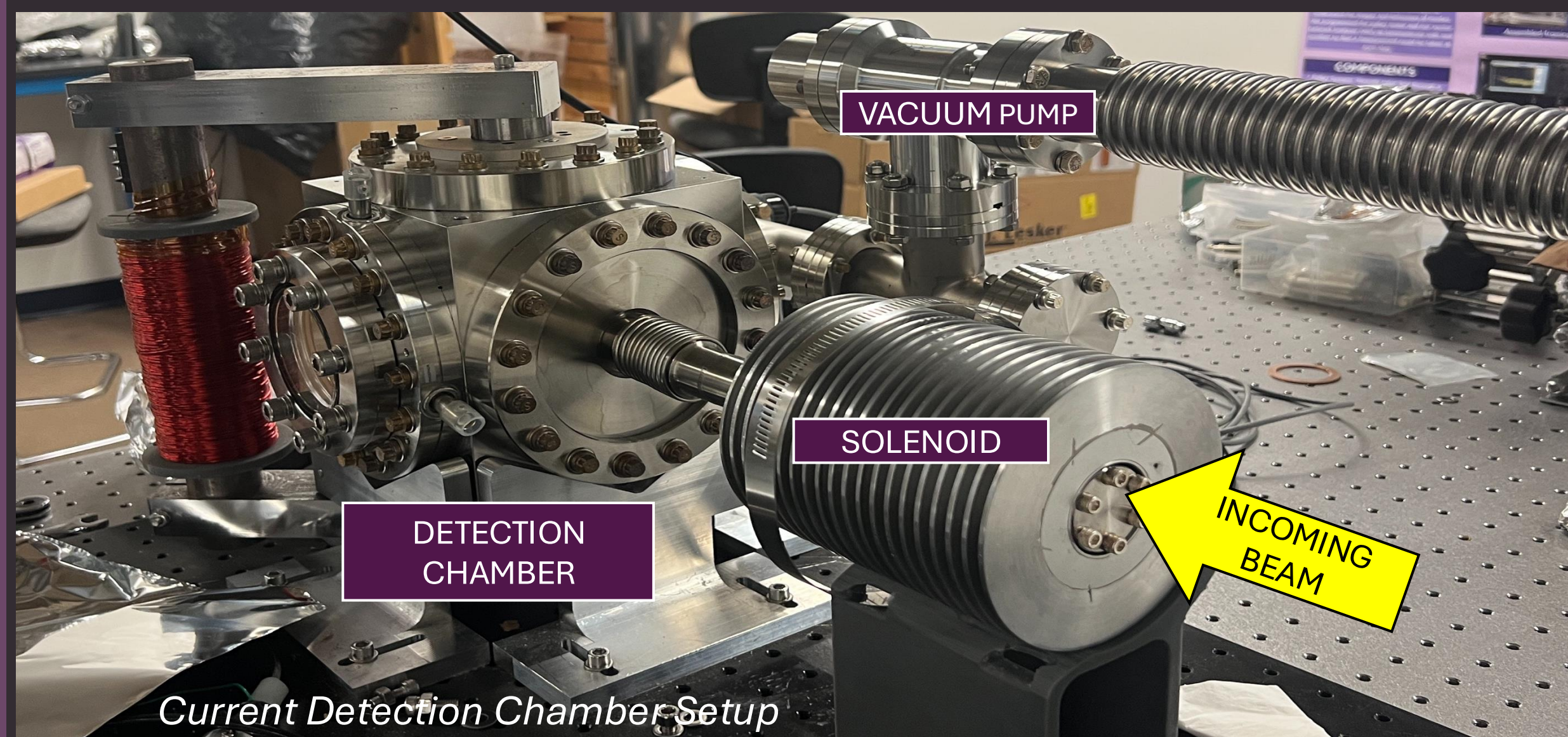
The electron gun uses thermionic emission [2]:

- Small applied voltage heats up lanthanum hexaboride (LaB_6) filament in cathode
- Electrons in cathode material become energized enough to overcome potential barrier, becoming free electrons that escape the surface of the material into the vacuum space

Our progress with electron gun testing falls under three stages:

- Controlled increase in source voltage while maintaining ultra high vacuum
- Achieve stable emitted current from the gun, as well as higher-resolution detected current reading of an on-axis Faraday cup connected to the end of the gun
- Currently continuing to correlate the energy, grid, and focus with detected current under the same source voltage, which requires finer grain and automatic data recording. Coco is working on programming for continuous data readout and real-time visualization.

Detection Chamber Setup



Current Detection Chamber Setup

Once the beam is accelerated in the four-cell linac, it passes through a solenoid that uses fringes of its static magnetic field to focus the beam in the direction of the detection chamber, which includes a dipole energy spectrometer, two micro-channel plates (MCPs), a Faraday cup, and two phosphor screens.

Dipole magnet energy spectrometer: Introduces a displacement to the incoming beam.

- Higher energy particles have a larger bending radius than lower energy particles.
- The narrow energy range requires precise field characterization.
- Directs the beam to the MCPs to further characterize its energy spread.

Faraday cup: Measures the current from an electron beam, which converts to the number of electrons.

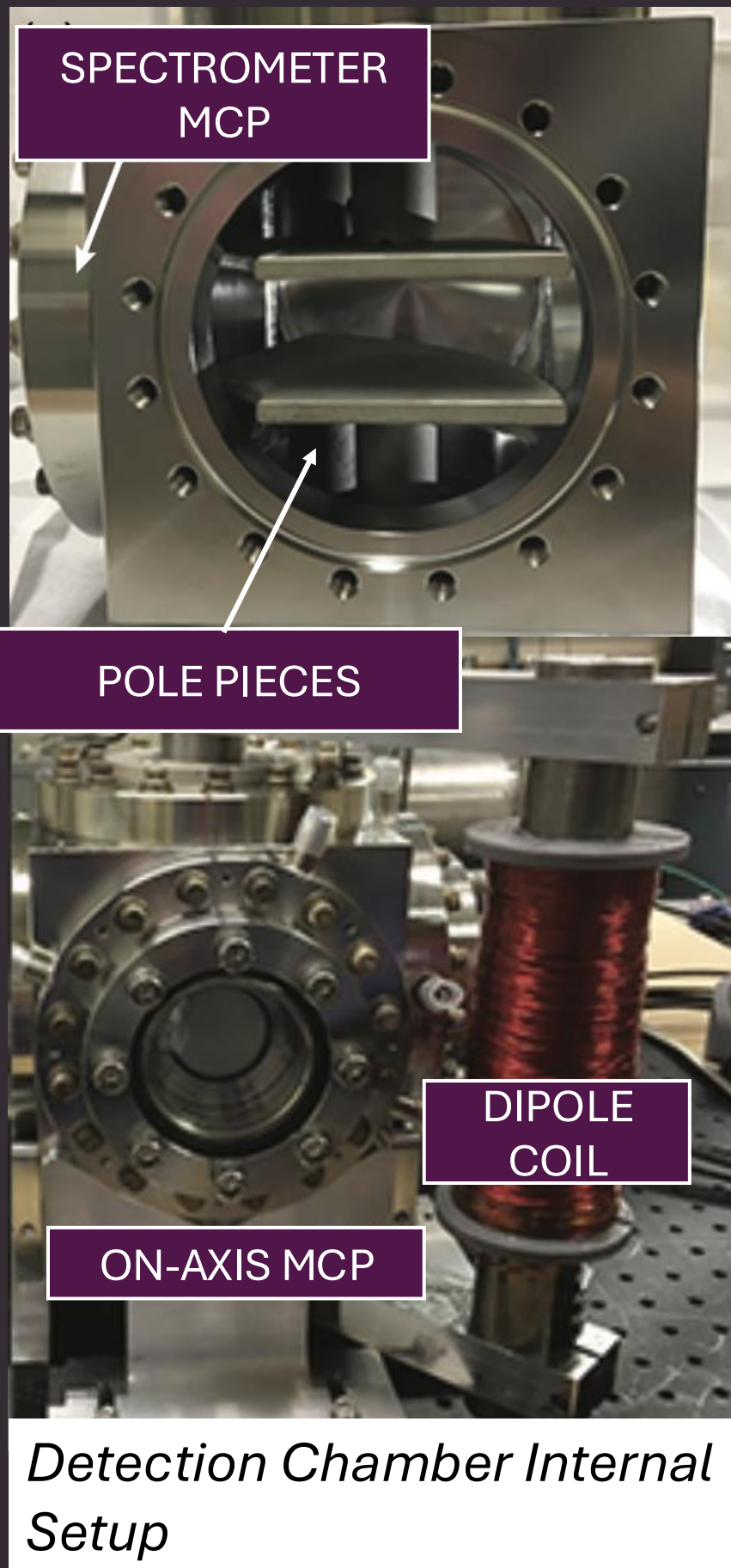
- The off-axis Faraday cup in the detection chamber can monitor dark current from breakdown or measure the incoming beam current when the dipole is reversed.

Micro-channel Plates (MCPs): Increases the beam signal by multiplying electrons.

Scintillating phosphorus screen: Turns the electrons to visible photons.

Camera: Captures the beam in frames

- Frames will be analyzed and converted to physical measurements of the energy spread and intensity.



Detection Chamber Internal Setup

Accelerating Cavities

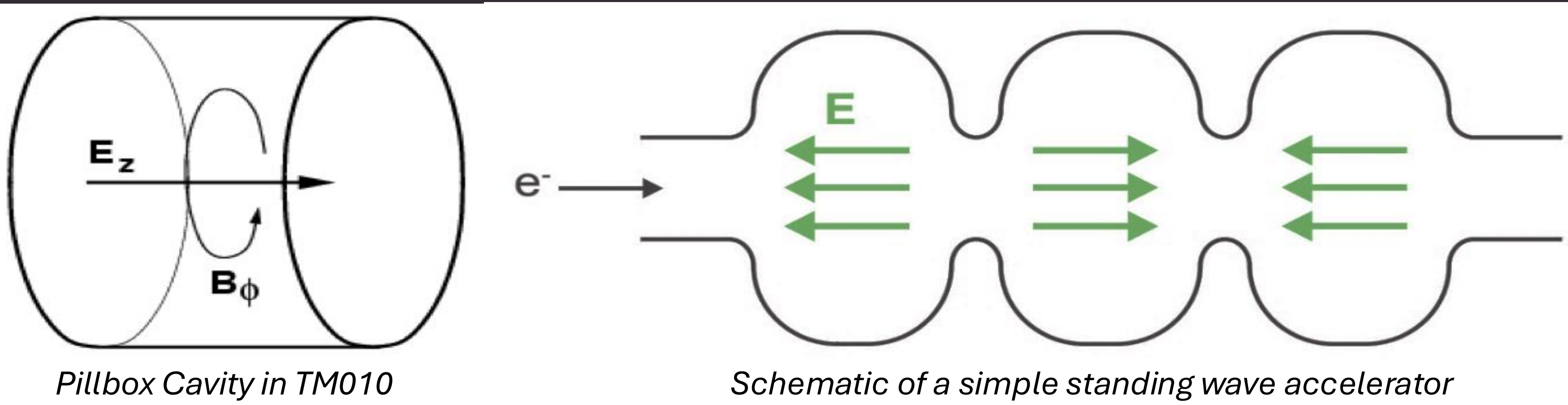
The resonant cavities used in this configuration are cylindrical 'pillbox' cavities:

- Physical dimensions of the cavity enable discrete resonant frequencies of standing wave, known as *eigenmodes*, which provide EM solutions to the wave equation
- To fulfill the metal boundary condition at the cavity walls, E field is strictly perpendicular to the surface and B-field is strictly parallel to the surface
- This setup utilizes the lowest frequency solution in transverse magnetic (TM), mode TM₀₁₀. Transverse magnetic indicates the magnetic field is perpendicular to the EM wave's direction of propagation.
- In TM₀₁₀ mode, E-field is uniform along the z-axis, which is our beam axis, and B-field is in azimuthal symmetry.

Generally, the kinetic energy gained by a particle of charge e is

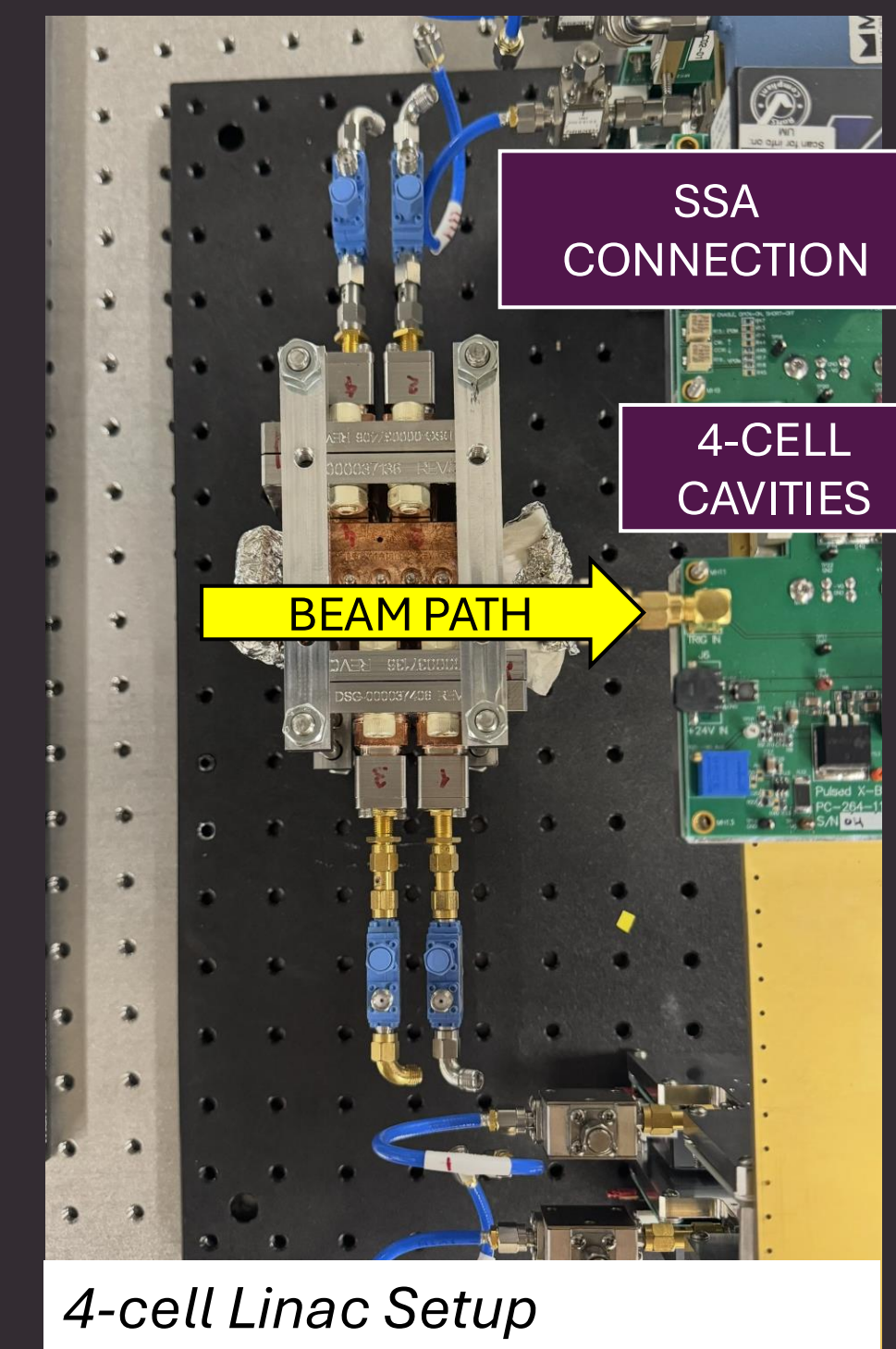
$$\Delta E = e\Delta V,$$

where ΔV is the potential difference that the particle is accelerating through, which in each cavity is determined by the time-varying on-axis E-field strength over length.



Pillbox Cavity in TM₀₁₀

Schematic of a simple standing wave accelerator



4-cell Linac Setup

Our setup couples four accelerating cavities. The phase between cells is in 0-mode, where all cells are in phase with each other. This means that while the field oscillates, the E field is always in the same direction at any time. The field is timed so that the electrons experience a continuous acceleration. [1,2]

Each cavity is driven directly by a solid-state amplifier board, which is the main R&D effort from researchers and engineers at SLAC.

- The architecture in these solid-state amplifiers provides up to 100W of power per cavity at X-band frequencies, eliminating the need for klystrons. Also provide higher stability and resolution of beam [4]
- The Gallium Nitride high-electron mobility transistors (HEMT) in SSAs are compact and operate with low-voltage power supplies. [1]

FUTURE

- Map the relation between gun control parameters and current
- Measure an energy spread by connecting the gun to the detection chamber
- Characterize beam and energy spread using image analysis program
- Characterize the B field of dipole magnet and compare it with the numerical analysis result from Ansys Maxwell
- Characterize the four-cell acceleration cavities from SLAC and eventually integrate it into the full system
- Integrate a specimen structure to use the accelerator for electron microscopy

REFERENCES

- [1] J. Sherman, Commissioning a Solid State Driven Linac for Microcrystal Electron Diffraction. BA thesis, Wellesley College 2025.
- [2] S. Lewis, A THz-Driven Electron Gun. PhD thesis, University of California, Berkeley 2020
- [3] J. Sherman, et. al., Solid State Driven X-band Linac for Microcrystal Electron Diffraction. IPAC Proceeding 2025.
- [4] A. Dhar, et. al. Solid State Driven Linac for Electron Microscopy. IPAC Proceeding 2024.
- [5] A. Dhar, et. al. Solid State Driven Linac for Electron Microscopy. IPAC Proceeding 2021.